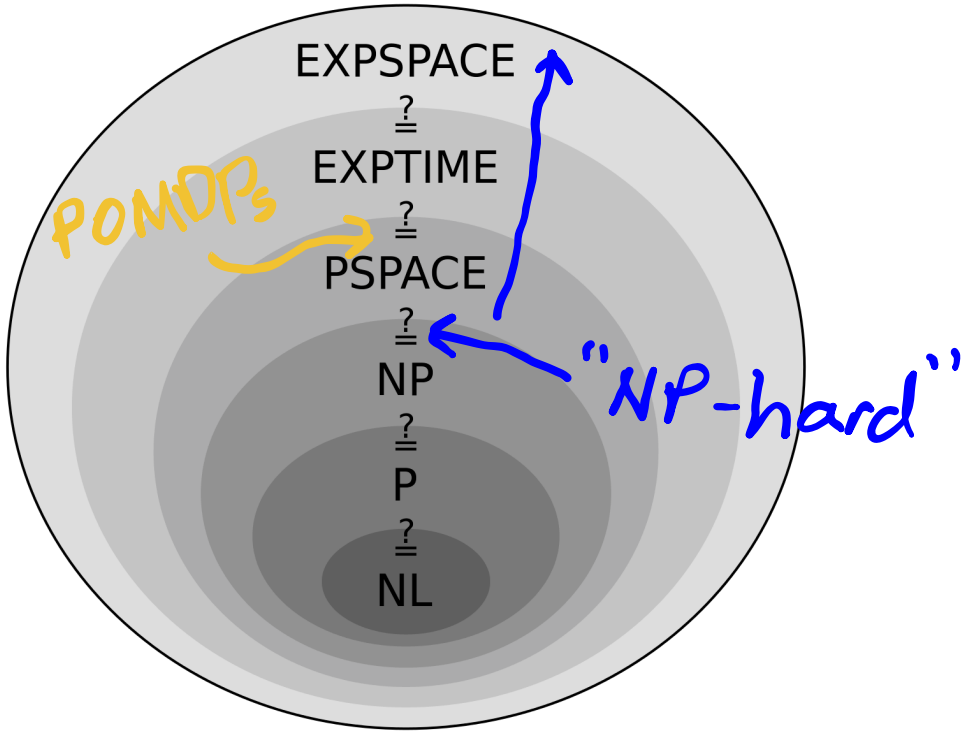
POMDP Formulation Approximations

POMDP Computational Complexity

Sad facts 😭

- Infinite horizon POMDPs are *undecidable*
- Finite horizon POMDPs are *PSPACE Complete*
  - Among the hardest problems that can be solved using a polynomial amount of space
  - Any algorithm that can solve a general POMDP will have exponential complexity

    (we think)



Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)

Offline

SARSOP

Online

(Next time)

Formulation Approximations

(solve a slightly different problem)

Today!

POMDP Objective

$$\pi^* = \underset{\pi:B\to A}{\operatorname{argmax}} \ \mathrm{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t))\right]$$

$$b' = \tau(b, a, o)$$

Certainty Equivalent

POMDP Objective

$$\pi^* = \underset{\pi:B\to A}{\operatorname{argmax}} \ \mathrm{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t))\right]$$

$$b' = \tau(b, a, o)$$

$$\pi_{\mathrm{CE}}(b) = \pi_s(\hat{s}(b))$$

$$b' = \tau(b, a, o)$$

Certainty Equivalent

Optimal for Linear-Quadratic-Gaussian (LQG)

(Analogous to LQR MDP)

<u>LQG POMDP</u>

$$S = \mathbb{R}^n, A = \mathbb{R}^m, O = \mathbb{R}^p$$

$$T(\mathbf{s}' \mid \mathbf{s}, \mathbf{a}) = \mathcal{N}(\mathbf{s}' \mid \mathbf{T}_s\mathbf{s} + \mathbf{T}_a\mathbf{a}, \mathbf{\Sigma}_s)$$

$$O(\mathbf{o} \mid \mathbf{s}') = \mathcal{N}(\mathbf{o} \mid \mathbf{O}_s\mathbf{s}', \mathbf{\Sigma}_o)$$

$$R(s, a) = -s^\top R_s s - a^\top R_a a$$

$$b(\mathbf{s}) = \mathcal{N}(\mathbf{s} \mid \mathbf{\mu}_b, \mathbf{\Sigma}_b)$$

$$\mathbf{\mu}_p \leftarrow \mathbf{T}_s\mathbf{\mu}_b + \mathbf{T}_a\mathbf{a}$$

$$\mathbf{\Sigma}_p \leftarrow \mathbf{T}_s\mathbf{\Sigma}_b\mathbf{T}_s^\top + \mathbf{\Sigma}_s$$

$$\mathbf{K} \leftarrow \mathbf{\Sigma}_p\mathbf{O}_s^\top\left(\mathbf{O}_s\mathbf{\Sigma}_p\mathbf{O}_s^\top + \mathbf{\Sigma}_o\right)^{-1}$$

$$\mathbf{\mu}_b \leftarrow \mathbf{\mu}_p + \mathbf{K}\left(\mathbf{o} - \mathbf{O}_s\mathbf{\mu}_p\right)$$

$$\mathbf{\Sigma}_b \leftarrow (\mathbf{I} - \mathbf{K}\mathbf{O}_s)\mathbf{\Sigma}_p$$

$$\pi^*_{\mathrm{LQG}}(b) = -K_{\mathrm{LQR}}\,\mu_b$$

QMDP

POMDP Objective

$$\pi^* = \underset{\pi:B\to A}{\operatorname{argmax}} \ \mathrm{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t))\right]$$

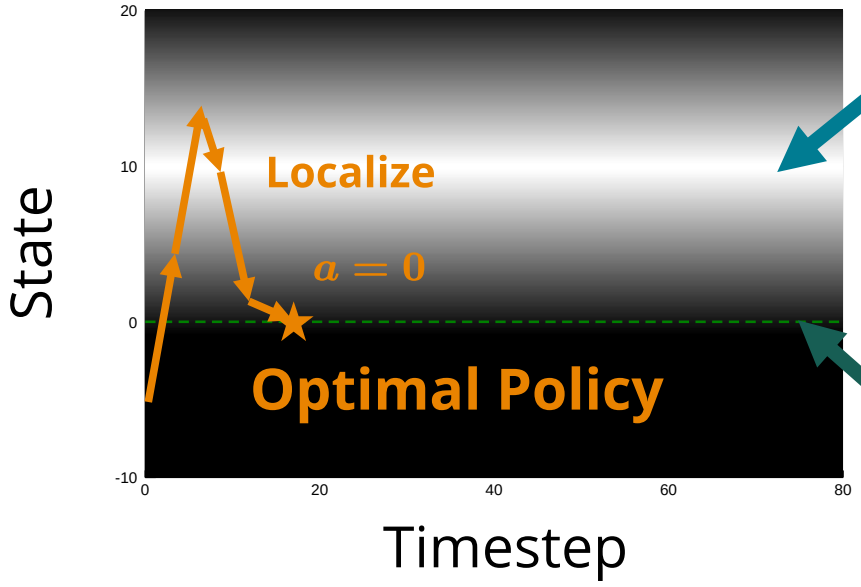
$$b' = \tau(b, a, o)$$

$$\pi_{\mathrm{QMDP}}(b) = \underset{a\in A}{\operatorname{argmax}} \ \underset{s\sim b}{\mathrm{E}}\left[Q_{\mathrm{MDP}}(s, a)\right]$$

$$b' = \tau(b, a, o)$$

Example: Tiger POMDP with Waiting

POMDP Example: Light-Dark



Accurate Observations

$$\mathcal{S} = \mathbb{Z} \qquad \mathcal{O} = \mathbb{R}$$

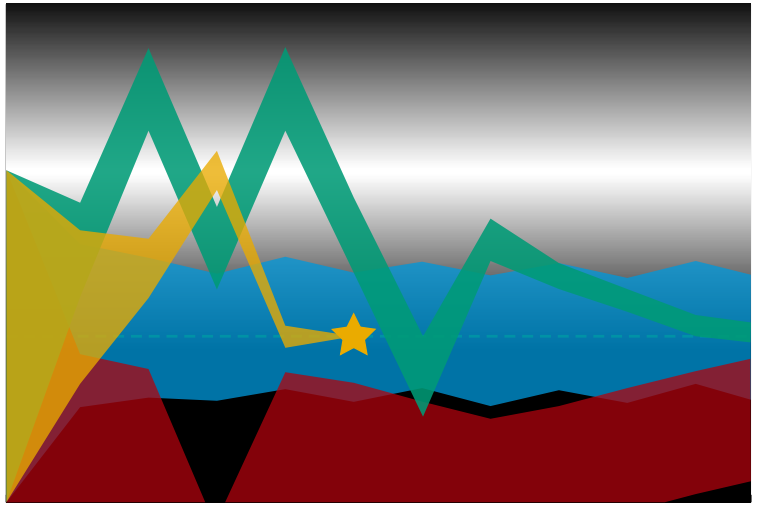
$$s' = s + a \qquad o \sim \mathcal{N}(s, s - 10)$$

$$\mathcal{A} = \{-10, -1, 0, 1, 10\}$$

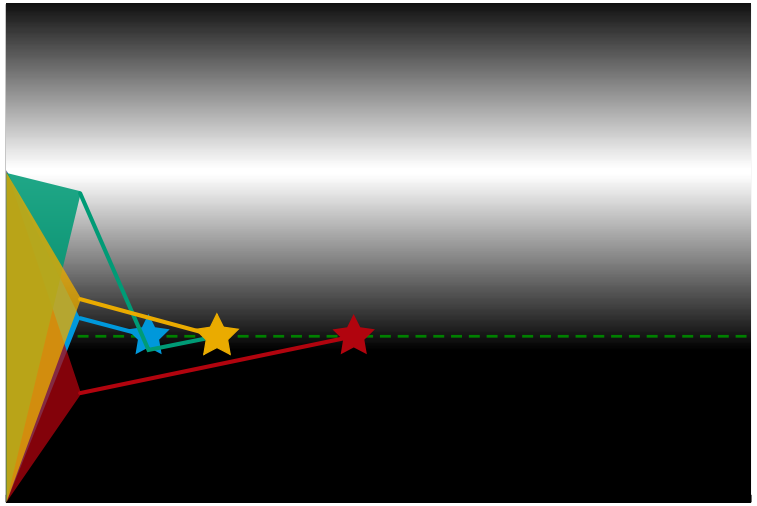
$$R(s,a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

Goal: $a = 0$ at $s = 0$

POMDP "Imagination"

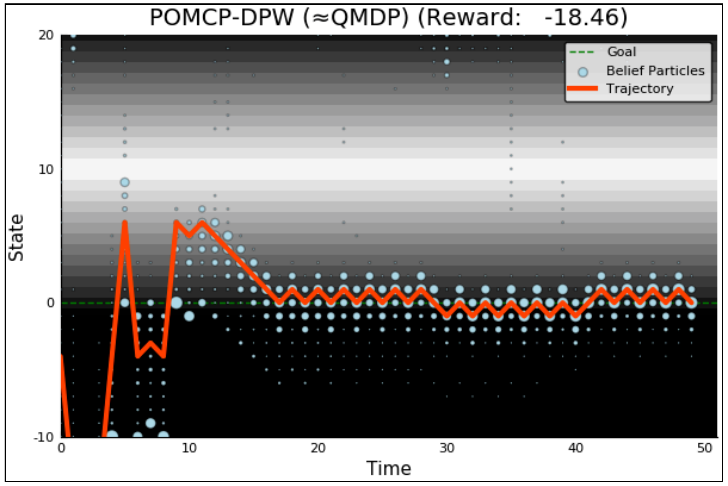


QMDP "Imagination"



QMDP Behavior

(Break)



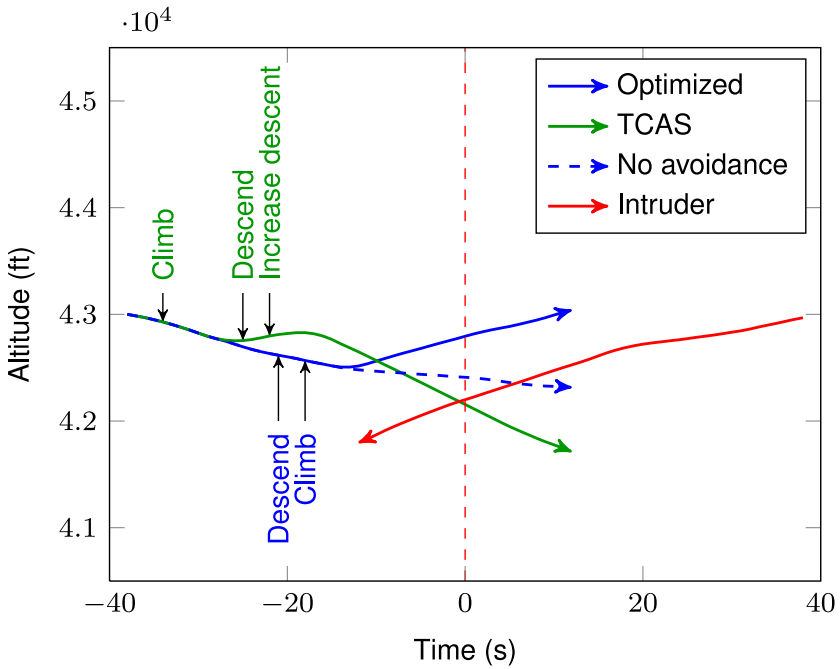
Same as **full observability** on the next step

QMDP

Very effective for most POMDPs and cheap (same cost as solving MDP)

Used for ACAS X

[Kochenderfer, 2011]



Two cases where it may not work well:

- *Costly* information gathering
- Long-lived uncertainty

Hindsight Optimization

POMDP Objective

$$\pi^* = \underset{\pi:B\to A}{\operatorname{argmax}} \ \mathrm{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t))\right]$$

$$b' = \tau(b, a, o)$$

k-Markov

POMDP Objective

$$\pi^* = \underset{\pi:B\to A}{\operatorname{argmax}} \ \mathrm{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t))\right]$$

$$b' = \tau(b, a, o)$$

Open Loop

A.K.A. Open Loop Feedback Control (OLFC)

POMDP Objective

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$$b' = \tau(b, a, o)$$